

CHE 330 HW 1 Questions

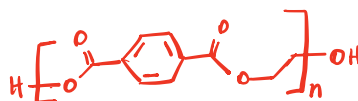
Assigned 8/27/2025, Due 9/02/2025 11:59 PM

Polymers Basics

1. Read Chapter 1 of the course textbook ([Paul C. Hiemenz, Timothy P. Lodge - Polymer Chemistry-CRC Press \(2007\).pdf](#)) and the reading/viewing assignments that were posted on the UIC blackboard.

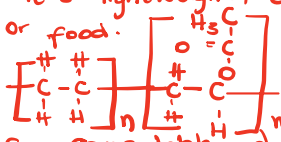
2. List five polymers that you can find in your daily life.
 - a. For each polymer, please provide the name of the polymer, the chemical structure of the monomer, and the chemical structure of the polymer using the right notation.
 - b. Explain why it is used for that application.

Water bottles: Polyethylene terephthalate
(PET) is used for water bottles because it's lightweight, strong, clear, and does not react with water or food.

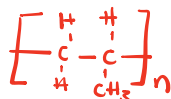


Shoes: Ethylene-vinyl acetate (EVA)

EVA is a versatile, foamy plastic used for comfortable, durable and lightweight soles and midsoles.



Toothbrush: Polypropylene



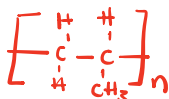
PP is cost-effective, thermoplastic, durable, light weight and resistant to chemicals like acids found in the mouth.

Clothes: Polyester



Polyester is durable, lightweight, affordable, wrinkle-resistant and moisture-wicking.

Hangers: Polypropylene



PP is used because of its strength and durability used in general-purpose and heavy-duty hangers.

3. Explain the difference between monomer, dimer, oligomer, polymer.

monomer - single unit
dimer - 2 units
oligomer - few units
polymer - many units



Polymer Classification

4. Would you consider a polymer of MW=25,000 a low polymer or a high polymer? Why?

High-MW polymers are greater than 20,000 Da. So a polymer of MW=25,000 is a high MW polymer.



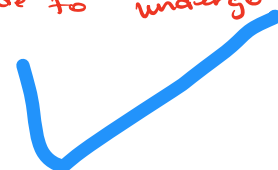
5. Identify five polymers that can be found in nature

- Cellulose
- Wool
- rubber
- DNA
- Chitin : genetic polymer composed of nucleotides



6. Natural polymers are made by living organisms, while synthetic polymers are produced in the lab by chemical reactions. Explain a semi-synthetic polymer.

Semi-synthetic polymers are modified natural polymers. They are derived from nature but are made to undergo chemical processes to enhance their quality.



7. List the 2 types of thermal responses of a polymer that we discussed in class. Describe them in your own words and list an example from class for each.

Thermoplastic: can be reshaped multiple times

Thermosetting: cannot be remelted (once formed its just as is)

A good example is a water bottle vs. car tires.

a water bottle can be remelted while car tires cannot.

8. Which has a slower conversion, chain growth or step growth formation? (Hint: Use the graph)

Step growth has a slower conversion

9. Briefly discuss 2 differences between a copolymer and a homopolymer.

Homopolymer have a single type of monomer and consist of single species of repeating units. While copolymer consists of two or more type of repeating units and have two or more types of monomers.

10. Why do cellulose and starch, both made from glucose, have different properties?

Describe how the differences in their structure at different levels (basic building blocks, chain arrangement, and overall 3D shape) affect properties like how well they dissolve in water, their crystallinity, and their rigidity.

Basic Building Blocks

- Cellulose: composed of β -D-glucose units linked by β -1,4-glycosidic bonds
- Starch: composed of α -D-glucose units linked by α -1,4-glycosidic bonds (amylose) and α -1,6-glycosidic bonds (amylopectin)

Chain Arrangement

- Cellulose Linear chains of glucose units. The β -1,4 linkages cause each glucose unit to flip making a straight unbranched chain

- Starch : Amylose is linear with α -1,4 linkages
Amylopectin is branched with both α -1,4
and α -1,6 linkages

Overall 3D shape

- Cellulose : Linear chains form hydrogen bonds creating rigid, crystalline microfibrils
- Starch: Amylose forms a helical structure
Amylopectin forms a branched, tree-like structure

Properties

• Cellulose

- insoluble in water
- highly crystalline
- very rigid and strong

• Starch

- More soluble in water
- less crystalline
- less rigid and more flexible